

Limitations & Assumptions

What this project is, and isn't

This reconstructs the LCR from the category-level subtotals each bank already discloses - it validates the aggregation logic (do the disclosed outflow categories sum to the disclosed total, does the 75% inflow cap apply correctly, does $HQLA \div \text{Net Cash Outflows}$ land on the disclosed ratio) rather than rebuilding HQLA from individual asset-level haircuts. That second thing isn't possible from public data: RBI's disclosure template gives a single Total HQLA figure, not the Level 1 / Level 2A / Level 2B breakdown that would be needed to rebuild it bottom-up. That breakdown exists only in supervisory reporting banks file with RBI directly, which isn't public. `hqla.py` validates the one number that is public, rather than pretending to a precision the source data doesn't support.

The HDFC Sep-2025 reconciliation break

9 of the 10 bank-quarters in this dataset reconcile to the ± 0.05 percentage point tolerance exactly - most land at a 0.00 gap. One doesn't: HDFC's September 2025 quarter, as restated in the comparative column of HDFC's own March 2026 filing. Total Outflows and Total Inflows both match the PDF's printed figures exactly; Net Cash Outflows doesn't follow from them by the standard formula - it's about Rs 41,955 crore lower than Outflows minus Inflows implies. I checked this against the raw extracted table cells directly (not just the downstream computation) to rule out a parsing bug before accepting it as real, and it holds. I don't have a confirmed explanation. The two most plausible ones - a restatement that updated the outflow/inflow lines without updating the net figure in the same pass, or a rounding/averaging methodology change between when Sep-2025 was originally filed and when it was reprinted as a comparative column eight months later - are both consistent with what's in the PDF, and I can't distinguish between them from public disclosure alone. It's reported as a break, not smoothed into the other nine.

The 75% inflow cap never binds in this dataset

For both banks, in every quarter, computed Total Cash Inflows sit well below 75% of computed Total Outflows - nowhere close to binding. That's arguably the expected case for well-capitalized retail-funded banks (a bank with a large inflow base relative to its outflows would look aggressive on the LCR formula's own terms), but it also means this project's cap logic is validated on the "cap doesn't matter" case in every single observation. Whether the aggregation would still hold under stress - a bank where the cap actually binds - is untested here.

Secured wholesale funding has no disclosed raw balance

Row 4 (Secured wholesale funding) never has an unweighted figure in either bank's disclosure, in any of the 10 bank-quarters. Only its weighted (post-haircut) value is given. The funding-concentration profile in `funding_concentration.py` is built from retail deposits + unsecured wholesale funding only, with secured wholesale funding excluded rather than estimated. In practice this is a small omission next to the rest of the funding base - row 4's weighted values run Rs 4,877-11,046 crore across HDFC's five quarters against a funding base of Rs 23-27 lakh crore, and IndusInd discloses Rs 0 for this row in every quarter - but it's excluded because there's no unweighted balance to include, not because it's been checked and found negligible in unweighted terms (the weighted, post-haircut figure isn't the same number).

HDFC and IndusInd disclose in different currency units

HDFC's PDFs are denominated in Rs million; IndusInd's are in Rs crore. Every ratio computed in this project is scale-invariant within one bank's own figures, so this never affected a reconciliation or a cross-bank percentage comparison. It does mean the raw unweighted/weighted columns in `lcr_disclosure_lines.csv` are not directly comparable in absolute rupee terms across the two banks without conversion - which is why `unweighted_inr_crore` and `weighted_inr_crore` columns exist alongside them.

Two banks, five quarters

This isn't a system-wide or sector-wide claim. HDFC and IndusInd were picked for a genuine funding-profile contrast - one large and retail-deposit-heavy, one smaller with a much higher less-stable-deposit share - not as a random or representative sample of Indian banks. The comparative finding in `reports/liquidity_risk_memo.md` is a finding about these two banks over these five quarters, not a generalization to the banking system.

Quarterly, average-balance, backward-looking

The disclosed figures are quarterly averages (both banks' templates say so), not point-in-time or daily figures, and both intra-quarter volatility and any subsequent quarter's numbers are outside this dataset. There's no attempt here to model what either bank's LCR would do under a live stress scenario - that's a different kind of project (scenario design), not a reconciliation of what's already been disclosed.